

# SUFYAN SIDDIQUI

[sufyansiddiqui005@gmail.com](mailto:sufyansiddiqui005@gmail.com) | [+92-343-3353881](tel:+92-343-3353881) | [linkedin.com/in/sufyan-siddiqui005](https://linkedin.com/in/sufyan-siddiqui005) | [github.com/sufyan005](https://github.com/sufyan005)

## PROFESSIONAL SUMMARY

Computer Science undergraduate specializing in full-stack development and applied machine learning. Built a real-time multiplayer web app with WebSocket state sync and a production ML recommendation system serving 5,000+ items. Proficient in Python, Java, C++, and the TypeScript/React/Node.js ecosystem, with strong foundations in OOP, data structures, and algorithms.

## EDUCATION

|                                                                                               |                        |
|-----------------------------------------------------------------------------------------------|------------------------|
| <b>Bachelor of Science in Computer Science</b>                                                | 2023 – 2027 (Expected) |
| DHA Suffa University, Karachi, Pakistan   Currently in 7th Semester   <b>CGPA: 3.7 / 4.00</b> |                        |
| <b>Intermediate in Pre-Engineering</b>                                                        | 2023                   |
| PECHS Education Foundation Science College, Karachi                                           |                        |
| <b>Matriculation in Pre-Engineering</b>                                                       | 2021                   |
| Al Eman Education System, Karachi   Grade: A+   85.64%                                        |                        |

## TECHNICAL SKILLS

**Languages:** Python, Java, C++, C, JavaScript

**Web Development:** HTML5, CSS3, JavaScript, React.js, Responsive Design

**Data & ML:** Pandas, NumPy, SciPy, Scikit-learn, NLTK, TF-IDF, Cosine Similarity

**Databases:** SQL, MySQL, Database Design, Query Optimization

**Tools:** Claude Code, Codex, Git, GitHub, VS Code, Streamlit

**Core:** OOP, Data Structures, Algorithms, File Handling, System Design, Agile, SDLC

## PROJECTS

**Real-Time Multiplayer Mafia Game | TypeScript, React, Node.js, Express, Socket.io, Tailwind CSS, Zod, TanStack Query**

Live: [mafia-game-production-fb78.up.railway.app](https://mafia-game-production-fb78.up.railway.app)

- Engineered a scalable, full-stack real-time gaming application using React and Node.js, enabling seamless multiplayer interaction via a Socket.io event-driven architecture.
- Synchronized complex application state across distributed clients using robust WebSocket communication patterns for low-latency game logic, player actions, and live chat.
- Architected a responsive, component-driven frontend using React, Tailwind CSS, and Radix UI primitives for high accessibility and consistent UX across devices.

**Movie Recommender System | Python, Scikit-learn, Streamlit, TMDB API**

Live: [movie-recommender-for-all.streamlit.app](https://movie-recommender-for-all.streamlit.app)

- Built content-based filtering engine over 5,000+ movies using TF-IDF (5k features, bigrams) + cosine similarity, combining genre, cast, director, keywords, and plot into a unified feature vector.
- Achieved ~5x faster load times migrating from Pickle to Apache Parquet + SciPy NPZ, eliminating version-dependency errors across environments.
- Implemented parallel poster fetching via ThreadPoolExecutor against TMDB API with adjustable results (5–20) in a responsive Streamlit UI.

**Brick Breaker Game | Java**

- Engineered GUI-based arcade game using Java Swing/AWT with real-time rendering and collision detection (99% accuracy).
- Implemented game state management including scoring system, level progression, and game-over logic.

**Hangman Game | Java**

- Built interactive console word guessing game with 500+ word database, input validation, and complete game loop logic.

**Bookstore Management System | C**

- Developed inventory system with full CRUD operations using structures, arrays, and file handling for persistent storage.

## ADDITIONAL INFORMATION

**Languages:** English (Fluent), Urdu (Native)

**Interests:** Software Development, Web Technologies, AI/ML, Algorithm Design, UI/UX Design, Data Science